SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FaceForge: Multi-Modal Facial Emotion Recognition System (Image • Video • Real-Time Webcam)

Academic Year 2026-27 • Final Year Engineering Project • Collaborative 4-Member Team Structure

1. Document Control & Responsibility Allocation

Team Member	Module / Role	Core Technical Responsibility	Status in SRS
Member 1	Module 1: Preprocessing	Dataset collection (FER-2013), data partitioning, grayscale conversion, image resizing, and normalization.	**Template Stub** (To be filled by M1)
Member 2	Module 2: Computer Vision	OpenCV pipeline, face localization (x, y, w, h), ROI face cropping, multi-face handling, and video frame extraction.	**Template Stub** (To be filled by M2)
Member 3	Module 3: Emotion Model	Pretrained FER model integration, 7 emotion probabilities, evaluation metrics (F1, Precision, Recall), and confusion matrix.	**Template Stub** (To be filled by M3)
Member 4 (Active)	**Module 4: Integration & UI**	Full application entrypoint (app.py), REST API Gateway, light Web UI, CSV storage engine (results.csv), and analytics dashboard.	**100% Fully Built & Specified**

2. Project Overview & Scope

FaceForge is an engineering-grade computer vision and machine learning application developed to recognize human affective states from visual input. The system processes visual media, detects human faces, isolates facial regions of interest (ROI), and classifies emotions using an established pretrained deep learning model. The project is purposefully architected as a collaborative 4-part system so that each team member operates within a clearly decoupled module with strict input/output contracts.

Three Operating Modes Supported:

- **Mode 1 (Static Image):** Single-photo analysis with face bounding box overlays and confidence scores.
- **Mode 2 (Video File):** Offline video analysis with configurable frame sampling step to optimize computational throughput.
- **Mode 3 (Live Webcam):** Real-time browser camera stream inference via OpenCV/Webcam capture with live overlay telemetry.

3. Four-Layer System Architecture

Layer	Primary Function	Component Technologies	Owner
Layer 1: Input	Data acquisition from Image uploads, Video files, and live camera feed.	OpenCV VideoCapture, HTML5 File API, MediaDevices	Member 2 & 4
Layer 2: Vision	Preprocessing, face detection (bounding coordinates), and ROI extraction.	OpenCV, NumPy matrix slicing, Haar / Deep Detector	Member 1 & 2
Layer 3: AI Inference	Pretrained model inference returning 7 emotion probability vectors.	FER Library (Pretrained CNN), Softmax argmax	Member 3
Layer 4: Integration	Web UI, REST API Gateway, append-only CSV ledger, and analytics.	FastAPI, React (Light UI), pandas, matplotlib, storage.py	Member 4

4. Teammate Module Specifications (Modules 1, 2 & 3)

4.1 Module 1: Dataset & Preprocessing (Member 1)

[SECTION ALLOCATED TO MEMBER 1]
- ***Input:*** *Raw static images and FER-2013 directory partitions.*
- ***Output Contract:*** *Preprocessed normalized grayscale/RGB NumPy image array.*
- ***Tasks to complete:*** *Document dataset class distribution (Happy, Sad, Angry, etc.), resizing parameters, grayscale transformations, normalization equations, and test set partitioning in dataset_analysis.py.*

4.2 Module 2: Computer Vision & Face Detection (Member 2)

[SECTION ALLOCATED TO MEMBER 2]
- ***Input:*** *Image matrix or video stream frame.*
- ***Output Contract:*** *Face bounding box coordinates (x, y, w, h) and extracted face ROI array image[y:y+h, x:x+w].*
- ***Tasks to complete:*** *Document face detection algorithms, multiple face loop handling, video frame sampling mechanics (frame_count % 10 == 0), and camera stream FPS stabilization in face_detection.py and video_processor.py.*

4.3 Module 3: Pretrained Emotion Model & Evaluation (Member 3)

[SECTION ALLOCATED TO MEMBER 3]
- ***Input:*** *Cropped face ROI image array.*
- ***Output Contract:*** *7-class probability dictionary and dominant emotion label with confidence scalar.*
- ***Tasks to complete:*** *Document pretrained FER deep learning model architecture, benchmark accuracy on test partitions, generate 7x7 confusion matrix (confusion_matrix.png), and calculate Precision, Recall, and F1-Scores in metrics_report.txt.*

5. Module 4: Integration, Web UI, Storage & Analytics (Full Implementation)

Module 4 represents the centralized integration hub and customer-facing software suite. It coordinates all pipeline stages, enforces non-crashing execution using graceful mock adapters, records all detections into persistent storage, and presents an enterprise-grade light-themed user interface.

5.1 RESTful API Gateway Specification (`app.py`)

Endpoint	Method	Input Parameters	Output Response / Schema
/api/health	GET	None	System status badge, active adapter mode, documentation URL.
/api/config	GET / POST	JSON payload with feature flags (demo_mode, use_module1..3, sampling_step)	Updated configuration parameters, saved to config.yaml.
/api/detect/image	POST	Multipart form file (JPG / PNG)	Bounding coordinates, detected emotions, confidence, and annotated preview.
/api/detect/video	POST	Multipart video file (MP4 / AVI), sampling_step integer	Total frames processed, emotion sequence timeline, per-frame detections.
/api/detect/webcam-frame	POST	Base64 image string or multipart frame	Live dominant emotion tag, confidence percentage, status pill.
/api/results	GET	Query filters: source, emotion, data_source	Filtered rows from results.csv, total record count.
/api/results/download	GET	None	Direct file download of results.csv ledger archive.
/api/results/clear	DELETE	None	Empties results.csv after user confirmation.
/api/analytics	GET	Query filter: data_source (all/live/demo)	Aggregated statistical counts, class distribution, mean confidence.
/api/evaluation	GET	None	Returns metrics_report.txt content and confusion_matrix.png base64.

5.2 Telemetry Storage Schema (`storage.py` → `results.csv`)

All predictions across Image, Video, and Webcam modes are automatically logged into an append-only persistent CSV file with the following column structure:

Field Name	Type	Description	Sample Value
timestamp	String	Timestamp of detection event	2026-10-31 19:39:24
source	String	Medium: image, video, or webcam	webcam
file_name	String	Uploaded file or camera stream index	camera_0
frame_number	Integer	Frame index in video sequence	140
face_id	Integer	Index of face in multi-face frame	0
emotion	String	Classified dominant emotion	Happy
confidence	Float	Normalized model probability [0, 1]	0.92
status	String	High-level status tag	Happy
data_source	String	Provenance: live or demo mock	live

5.3 Light-Themed Web User Interface (`frontend/src/`)

- **Design System:** Clean Slate light theme (`#f8fafc` canvas, `#ffffff` cards, `#0f172a` text, `#4f46e5` electric indigo primary).
- **Home View (`HomeView.jsx`):** System architecture summary, active pipeline status pill, and direct one-click mode launches.
- **Image Detection View (`ImageView.jsx`):** Drag-and-drop file upload, preview rendering, and bounding box results table.
- **Video Detection View (`VideoView.jsx`):** HTML5 video player, interactive step-sampling range slider, and frame progress bar.
- **Webcam Detection View (`WebcamView.jsx`):** Camera stream with start/stop triggers and floating real-time emotion telemetry badge.
- **Results Ledger (`ResultsView.jsx`):** Tabular display with multi-level filtering (source, emotion, data tier) and CSV export.
- **Analytics Dashboard (`AnalyticsView.jsx`):** Statistical frequency chart, valence spread, mean confidence, and PDF summary.
- **Settings View (`SettingsView.jsx`):** Independent toggles for teammate adapters (Modules 1, 2, and 3) and threshold controls.
- **Model Evaluation View (`EvaluationView.jsx`):** Displays Member 3's confusion matrix and parsed F1/Recall metrics report.

6. Master Verification & Testing Suite

Test ID	Mode	Input Condition	Expected System Outcome	Lead
T1	Image	Clear portrait with smile	Face detected, bounded, tagged Happy with >75% confidence.	M4
T2	Image	Sad portrait	Classified Sad, entry appended to results.csv.	M4
T3	Image	Angry expression	Classified Angry, probability distribution updated.	M4
T4	Image	Neutral face	Classified Neutral, displayed in results table.	M4
T5	Image	Multiple faces in frame	All distinct faces bounded and classified individually.	M2/M4
T6	Image	Scenery without human faces	Reports 0 faces detected without throwing server error.	M4
T7	Image	Dark / low-lighting image	System triggers low-confidence warning badge gracefully.	M1/M4
T8	Video	10-second MP4 clip	Frames extracted according to step N, progress bar renders.	M4
T9	Video	Multi-person video	Tracks detections across sampled frames sequentially.	M2/M4
T10	Video	Subject changing emotions	Analytics timeline shows temporal emotional transitions.	M3/M4
T11	Video	Low-resource hardware (i3)	Frame sampling skips intermediate frames, preventing lag.	M4
T12	Webcam	Normal ambient lighting	Real-time live video feed with floating prediction HUD.	M4
T13	Webcam	Dim ambient room	Video feed continues without camera disconnect.	M4
T14	Webcam	Subject turns away from camera	HUD shows No Face Detected gracefully without crash.	M4
T15	Storage	Reset database action	Requires modal confirmation, then safely clears results.csv.	M4

7. Limitations & Conclusion

Limitations: Processing speed is bounded by CPU clock frequencies when analyzing videos at 100% frame density. The system mitigates this via frame sampling steps. Furthermore, deep shadows or extreme camera angles reduce face localization accuracy.

Conclusion: The FaceForge project establishes a robust modular architecture for affective computing. By clearly demarcating responsibilities into 4 modules, Members 1, 2, and 3 are provided standardized interfaces to plug in their datasets, detectors, and evaluation models, while Member 4 provides the operational, scalable web integration, persistent ledger, and visualization engine.